

and normally-open (N/O) positions. The receiver takes 5V. When a signal is sent, it switches between N/C and N/O.

Software design

Autonomous challenge algorithm. Team Moksha adopted an ingenious approach for autonomy of the robot. The software design flow-chart is

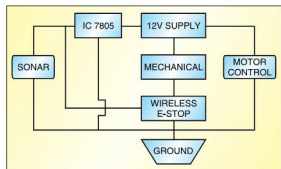


Fig. 3: Block diagram of the UGV's electrical system design

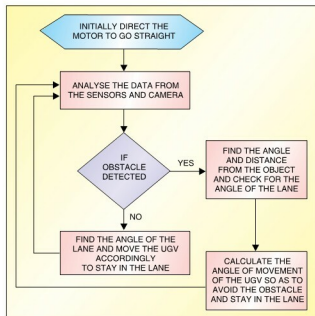


Fig. 4: Software design flow-chart

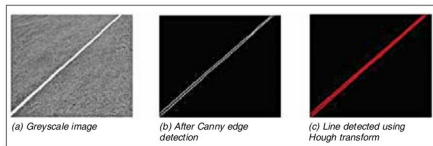


Fig. 5: Image processing

shown in Fig. 4.

Platform. The different devices on the UGV are controlled by a C++ program, which includes an integrated algorithm. The algorithm uses data from the sensing devices, camera, LIDAR and SONAR, and directs the RS160D motor controller. The different devices are connected to the processor using the USB ports, which are inter-

faced to the devices using USB-to-RS232 converters. This is done since the data is exchanged between the processor and the sensors and motor controller serially.

Image processing. The images are continuously taken from the video captured by a 5-megapixel camera and processed using different image-processing algorithms to detect the lane. This is done using the functions in the OpenCV library. The angle of the lane is calculated using the coordinates of the detected lines from the processed images and the instructions sent to the motor controller to move the UGV in the directed angle.

The camera is placed at a height of about 90 cm (3 feet) at an angle facing the ground in front

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of the vehicle. This is done to avoid detection of any unnecessary line due to noise in the image. Also, proper calibration is done in the program to match the angle of the lane to the angle of the line that appears in the image. The image is processed as follows (refer Fig. 5):

1. First, the RGB image is converted into greyscale.

2. The greyscale image is processed using Canny algorithm to detect the edges.

3. The resulting image is then processed using Hough (Probabilistic) transform to detect only line segments in the image.

Obstacle detection. A serial communication program reads the values continuously from the LIDAR through the serial port. The values are in hex format, which are converted into integer. The protocol structure consists of the following bytes that may be received: start, operation, option (1, 2, 3), scan number, angular step size, start angle, stop angle and distance.

Angular step size specifies the angular separation between two successive transferred measurement values and is equal to 0.36 degree. The difference between start and stop angles gives an approximation about the width of the obstacle. The difference is multiplied by the angular step size to calculate the width of the obstacle. The distance bit is the measured distance of an obstacle from the current location of the LIDAR. Using the distance, start and stop bits, the area around the UGV can be mapped. After knowing the distance and the angle of the obstacle,